Reference guide: Linux

Google Cybersecurity Certificate

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Navigate the file system

The following Linux commands are helpful when navigating the file system.

cd

Navigates between directories

> `cd reports`
> Navigates from the current working directory to its subdirectory `reports`
>
> `cd /home/analyst/reports`
> Navigates to the `reports` directory; the full path is required when `reports` is not a subdirectory of the current working directory
>
> `cd ..`
> Navigates to the directory that is one level above the current working directory

ls

Displays the names of the files and directories

`ls`
Displays the names of the files and directories in the current working directory

`ls /home/analyst/reports`
Displays the names of the files and directories in the `reports` directory; providing an argument that specifies the path to a directory is necessary to display the contents of a directory other than the user's current working directory

`ls -a`
Displays hidden files when displaying the names of files and directories inside the current working directory

`ls -l`
Displays permissions to files and directories in the current working directory; also displays other additional information, including owner name, group, file size, and the time of the last modification

`ls -la`
Displays permissions to files and directories in the current working directory, including hidden files; also displays other additional information, including owner name, group, file size, and the time of last modification

pwd

Prints the working directory to the screen

`pwd`
Prints the working directory to the screen, such as `/home/analyst`

whoami

Returns the username of the current user

`whoami`
Returns the username of the current user, such as `analyst` or `fgarcia`

Read files

The following Linux commands are helpful when reading files.

cat

Displays the content of a file

> `cat updates.txt`
> Displays the content of the `updates.txt` file

head

Displays just the beginning of a file, by default 10 lines

> `head updates.txt`
> Displays only the first 10 lines of the `updates.txt` file

> `head -n 5 updates.txt`
> Displays only the first five lines of the `updates.txt` file; the `-n` option allows users to specify the number of lines to return

less

Returns the content of a file one page at a time

> `less updates.txt`
> Returns the content of `updates.txt` one page at a time; the `less` command changes the terminal window to a display that allows users to easily move forward and backward through content

tail

Displays just the end of a file, by default 10 lines

> `tail updates.txt`
> Displays only the last 10 lines of the `updates.txt` file

`tail -n 5 updates.txt`
returns only the last five lines of the `updates.txt` file; the `-n` option allows users to specify the number of lines to return

Manage the file system

The following Linux commands are helpful when managing the file system.

cp

Copies a file or directory into a new location; the file will not be removed from the previous location

`cp permissions.txt /home/analyst/logs`
Copies the `permissions.txt` file from the user's current working directory to the `logs` directory

mkdir

Creates a new directory

`mkdir network`
Creates a new directory named `network` in the user's current working directory

`mkdir /home/analyst/logs network`
Creates a new directory named `network` in the `logs` directory; the full path is required when `logs` is not a subdirectory of the current directory

mv

Moves a file or directory to a new location; the file is also removed from the previous location

`mv permissions.txt /home/analyst/logs`
Moves the `permissions.txt` file from the user's current working directory to the `logs` directory

`mv permissions.txt perm.txt`
Moves the `permissions.txt` file from the user's current working directory to the new file name `perm.txt` in the user's current working directory; this results in renaming the `permissions.txt` file as `perm.txt`

nano

Opens or creates a file in the nano command-line file editor

`nano permissions.txt`
Opens an existing `permissions.txt` file in the nano file editor, or creates the `permissions.txt` file in the nano file editor if it doesn't already exist in the current working directory

rm

Removes, or deletes, a file

`rm permissions.txt`
removes the `permissions.txt` file from the user's current working directory

`rm home/analyst/reports/permissions.txt`
Removes the `permissions.txt` file from from the `reports` directory; the full path is required if the user's current working directory is not `reports`

rmdir

Removes, or deletes, a directory; only removes directories if they are empty

`rmdir network`
Removes the empty `network` subdirectory of the user's current working directory from the file system

`rmdir /home/analyst/logs/network`
Removes the empty `network` directory from the file system; the full path is required when `network` is not a subdirectory of the current directory

touch

Creates a new file

> `touch permissions.txt`
> Creates a new file named `permissions.txt` in the user's current working directory
>
> `touch /home/analyst/reports/permissions.txt`
> Creates a new file named `permissions.txt` in the `reports` directory; the full path is required if the user wants to create `permissions.txt` in any directory other than the current working directory

Filter content

The following Linux commands are helpful when filtering content.

find

Searches for directories and files that meet specified criteria

> `find /home/analyst/projects`
> Searches for all files starting at the `projects` directory
>
> `find /home/analyst/projects -name "*log*"`
> Searches for all files in the `projects` directory that contain the word `log` in the file name; the `-name` option searches for a specified string and is case-sensitive; the `*` wildcard represents zero or more unknown characters
>
> `find /home/analyst/projects -iname "*log*"`
> Searches for all files in the `projects` directory that contain the word `log` in the file name; the `-iname` option searches for a specified string and is not case-sensitive; the `*` wildcard represents zero or more unknown characters
>
> `find /home/analyst/projects -mtime -3`
> Searches for all files in the `projects` directory that have been modified within the past three days; the `-mtime` option bases its search for files or directories that were modified on days

`find /home/analyst/projects -mmin -15`
Searches for all files in the `projects` directory that have been modified within the past 15 minutes; the `-mmin` option bases its search for files or directories that were modified on minutes

grep

Searches a specified file and returns all lines in the file containing a specified string

`grep OS updates.txt`
Searches the `updates.txt` file and returns all lines containing the string `OS`

| (piping)

Sends the standard output of one command as standard input to another command for further processing; accessed using the pipe character (`|`)

`ls /home/analyst/reports | grep users`
Redirects the standard output of `ls /home/analyst/reports` to be standard input for the `grep users` command, meaning that `grep users` identifies files and subdirectories in the `/home/analyst/reports` directory that contain the string `users` within their file name

Manage users and their permissions

The following Linux commands are helpful when managing user permissions. (Also review the subentries for `ls -l` and `ls -la` in the `ls` entry of the **Navigate the file system** section.)

chmod

Changes permissions on files and directories

`chmod u+rwx,g+rwx,o+rwx login_sessions.txt`
Changes user (`u`), group (`g`), and other (`o`) permissions to add (`+`) read (`r`), write (`w`), and execute (`x`) permissions for the `login_sessions.txt` file

`chmod g-rw bonuses.txt`
Changes the group (`g`) permissions to remove (`-`) read (`r`) and write (`w`) permissions for the `bonuses.txt` file

`chmod u=r,g=r,o=r login_sessions.txt`

Changes user (`u`), group (`g`), and other (`o`) permissions to assign (`=`) read (`r`) permissions for the `login_sessions.txt` file

chown

Changes ownership of a file or directory; used with `sudo`

`sudo chown fgarcia access.txt`
Changes the user owner of the `access.txt` file to `fgarcia`

`sudo chown :security access.txt`
Changes the group owner of `access.txt` to `security`; a colon (`:`) must be entered before the group name

groupdel

Deletes a group from the system; used with `sudo`

`sudo groupdel accounting`
Deletes `accounting` as a group

sudo

Temporarily grants elevated permissions to specific users; users must be in a sudoers file to use have access to `sudo`

`sudo useradd fgarcia`
Grants elevated permissions to the user running this command and so that this user can use the `useradd` command to add `fgarcia` as a new user to the system

useradd

Adds a user to the system; used with `sudo`

`sudo useradd fgarcia`
Adds `fgarcia` as a new user to the system

`sudo useradd -g security fgarcia`
Adds `fgarcia` as a new user and uses the `-g` option to set their primary group as `security`

`sudo useradd -G finance,admin fgarcia`
Adds `fgarcia` as a new user and uses the `-G` option to add them to the supplemental groups of `finance` and `admin`

userdel

Deletes a user from the system; used with `sudo`

`sudo userdel fgarcia`
Deletes `fgarcia` as a user

`sudo userdel -r fgarcia`
Deletes `fgarcia` as a user and deletes all files in their home directory

usermod

Modifies existing user accounts; used with `sudo`

`sudo usermod -g executive fgarcia`
Uses the `-g` option to change the existing `fgarcia` user's primary group to the `executive` group

`sudo usermod -G accounting fgarcia`
Uses the `-G` option to replace any supplemental groups the the existing `fgarcia` user is in with the supplemental `accounting` group; removes all other supplemental groups `fgarcia` is in

`sudo usermod -a -G marketing fgarcia`
Uses the `-a -G` options to add the existing `fgarcia` user to the supplemental `marketing` group; does not remove `fgarcia` from other supplemental groups

`sudo usermod -d /home/garcia_f fgarcia`
Uses the `-d` option to change the existing `fgarcia` user's home directory to `/home/garcia_f`

`sudo usermod -L fgarcia`
Uses the `-L` option to lock the existing `fgarcia` user's account so they cannot log in

`sudo usermod -l garcia_f fgarcia`
Uses the `-l` option to change the existing `fgarcia` user's login name to `garcia_f`

Get help in Linux

The following Linux commands are helpful when getting help in Linux.

apropos

Searches the manual page descriptions for a specified string

`apropos password`
Returns the manual pages of commands that contain the keyword `password`

`apropos -a graph editor`
Returns the manual pages of commands that contain both the keywords `graph` and `editor`; the `-a` option specifies to only return commands that contain all specified strings

man

Displays information on other commands and how they work; the output is called a “man page,” which is short for "manual page"

`man chown`
Displays detailed information about `chown` and how it works

whatis

Displays a description of a command on a single line

`whatis nano`
Displays the description of `nano` on a single line